In [1]:
```
import pandas as pd

# Load the CSV file (same folder as notebook)
df = pd.read_csv("Vacant_Properties.csv")

# Basic checks
print("Dataset shape:", df.shape)
print("\nColumn names:")
print(df.columns.tolist())

# Preview data
df.head()
```

```
Dataset shape: (1651, 17)

Column names:
['X', 'Y', 'SBL', 'PropertyAddress', 'Zip', 'Owner', 'OwnerAddress', 'Vacant', 'neig
hborhood', 'VPR_result', 'completion_date', 'completion_type_name', 'valid_until',
'VPR_valid', 'Latitude', 'Longitude', 'ObjectId']
```

Out[1]:

	X	Y	SBL	PropertyAddress	Zip	Owner	OwnerAddre
0	NaN	NaN	002.-04-02.4	1920 Park St	13208	Wellington Ward LLC	100 Windsor Syracuse, N 132
1	-8.476690e+06	5.315382e+06	077.-23-25.0	316 Warner Ave	13205	Donna Larode	321 Warn Ave Syracus NY 132
2	-8.476078e+06	5.315534e+06	077.-10-15.0	2223 State St S & Amherst Ave	13205	Vladimir Dyomin	2644 Torr Pines Rd Jolla, CA 920
3	-8.479473e+06	5.319570e+06	109.-05-04.0	353 Richmond Ave	13204	Jason Yagan	353 Richmor Ave Syracus NY 132
4	-8.475432e+06	5.320042e+06	030.-03-01.0	500 Hawley Ave & Crouse Ave N	13203	City of Syracuse	233 Washington Syracuse, N 132

In [2]:
```
df.isna().sum().sort_values(ascending=False)
```

Out[2]:
```
valid_until             1276
VPR_result              1276
completion_type_name    1276
completion_date         1276
VPR_valid               1276
X                         36
Y                         36
Latitude                  36
Longitude                 36
neighborhood              15
OwnerAddress               0
Vacant                     0
Zip                        0
SBL                        0
PropertyAddress            0
Owner                      0
ObjectId                   0
dtype: int64
```

In [3]:
```
(df.isna().mean() * 100).round(2).sort_values(ascending=False)
```

Out[3]:
```
valid_until             77.29
VPR_result              77.29
completion_type_name    77.29
completion_date         77.29
VPR_valid               77.29
X                        2.18
Y                        2.18
Latitude                 2.18
Longitude                2.18
neighborhood             0.91
OwnerAddress             0.00
Vacant                   0.00
Zip                      0.00
SBL                      0.00
PropertyAddress          0.00
Owner                    0.00
ObjectId                 0.00
dtype: float64
```

In [4]:
```
null_summary = pd.DataFrame({
    "Missing_Count": df.isna().sum(),
    "Missing_Percent": (df.isna().mean() * 100).round(2)
}).sort_values("Missing_Count", ascending=False)

null_summary
```

Out[4]:

	Missing_Count	Missing_Percent
valid_until	1276	77.29
VPR_result	1276	77.29
completion_type_name	1276	77.29
completion_date	1276	77.29
VPR_valid	1276	77.29
X	36	2.18
Y	36	2.18
Latitude	36	2.18
Longitude	36	2.18
neighborhood	15	0.91
OwnerAddress	0	0.00
Vacant	0	0.00
Zip	0	0.00
SBL	0	0.00
PropertyAddress	0	0.00
Owner	0	0.00
ObjectId	0	0.00

In [5]:
```
data_dict = pd.DataFrame({
    "Column": df.columns,
    "Data Type": df.dtypes,
    "Missing Values": df.isna().sum()
})

data_dict
```

Out[5]:

	Column	Data Type	Missing Values
X	X	float64	36
Y	Y	float64	36
SBL	SBL	object	0
PropertyAddress	PropertyAddress	object	0
Zip	Zip	int64	0
Owner	Owner	object	0
OwnerAddress	OwnerAddress	object	0
Vacant	Vacant	object	0
neighborhood	neighborhood	object	15
VPR_result	VPR_result	object	1276
completion_date	completion_date	object	1276
completion_type_name	completion_type_name	object	1276
valid_until	valid_until	object	1276
VPR_valid	VPR_valid	object	1276
Latitude	Latitude	float64	36
Longitude	Longitude	float64	36
ObjectId	ObjectId	int64	0

In [6]:
```
analysis_df = df.drop(columns=[
    "VPR_result",
    "VPR_valid",
    "completion_date",
    "completion_type_name",
    "valid_until"
])
```

In [7]:
```
geo_df = analysis_df.dropna(subset=["Latitude", "Longitude"])
```

In [8]:
```
count_df = analysis_df.copy()
```

In [9]:
```
neighborhood_counts = count_df["neighborhood"].value_counts()
neighborhood_counts.head(10)
```

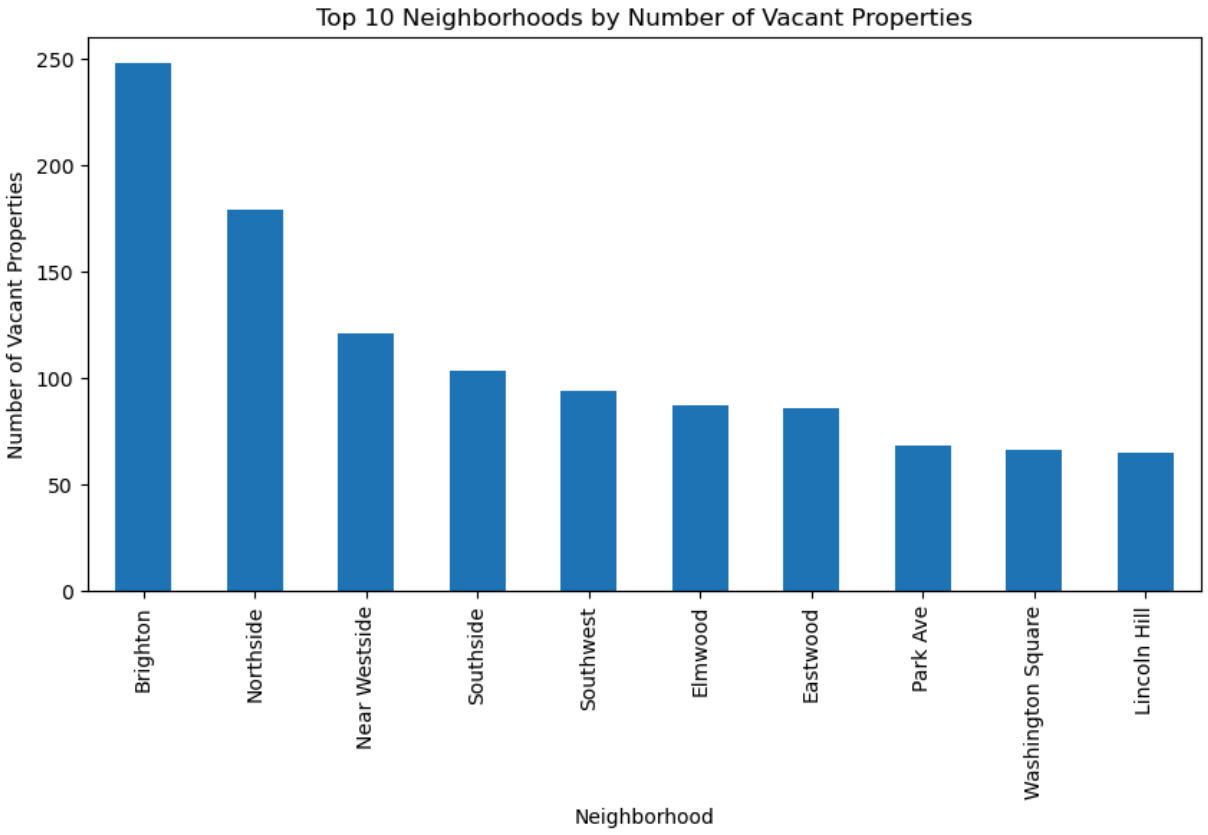
Out[9]:
```
neighborhood
Brighton             248
Northside            179
Near Westside        121
Southside            103
Southwest             94
Elmwood               87
Eastwood              86
Park Ave              68
Washington Square     66
Lincoln Hill          65
Name: count, dtype: int64
```

In [10]:
```
zip_counts = count_df["Zip"].value_counts()
zip_counts.head(10)
```

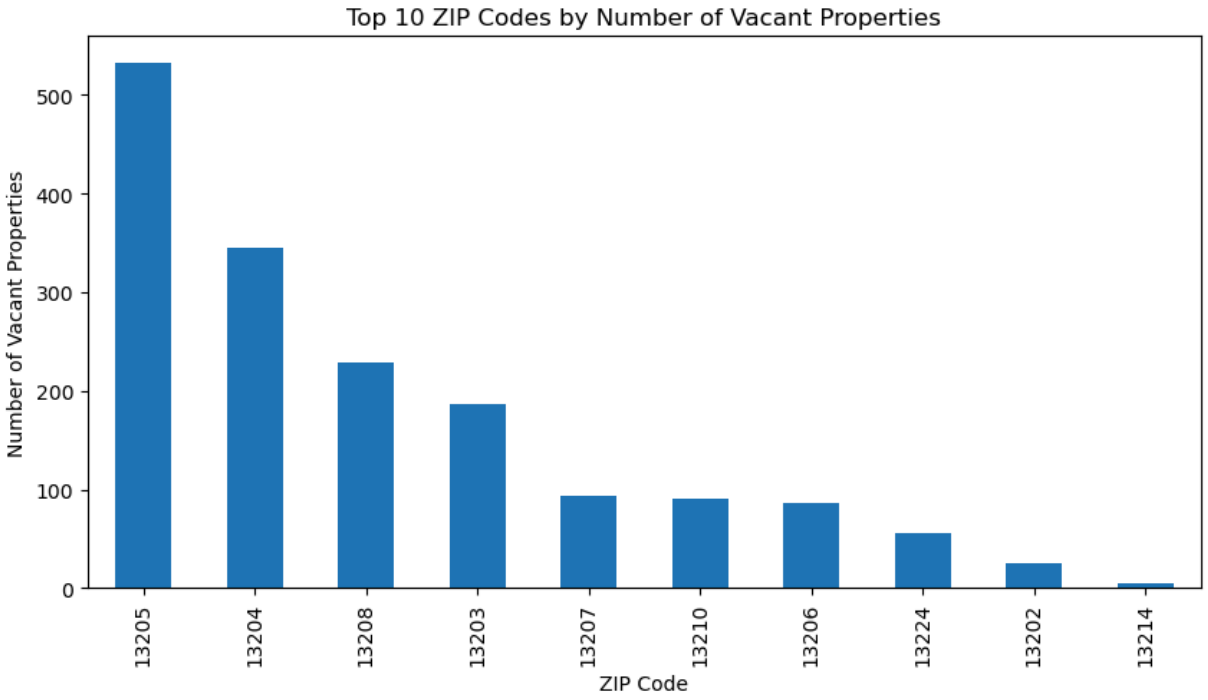
Out[10]:
```
Zip
13205    533
13204    345
13208    228
13203    186
13207     94
13210     91
13206     86
13224     56
13202     25
13214      5
Name: count, dtype: int64
```

In [11]:
```
import matplotlib.pyplot as plt

neighborhood_counts.head(10).plot(kind="bar", figsize=(10,5))
plt.title("Top 10 Neighborhoods by Number of Vacant Properties")
plt.xlabel("Neighborhood")
plt.ylabel("Number of Vacant Properties")
plt.show()
```

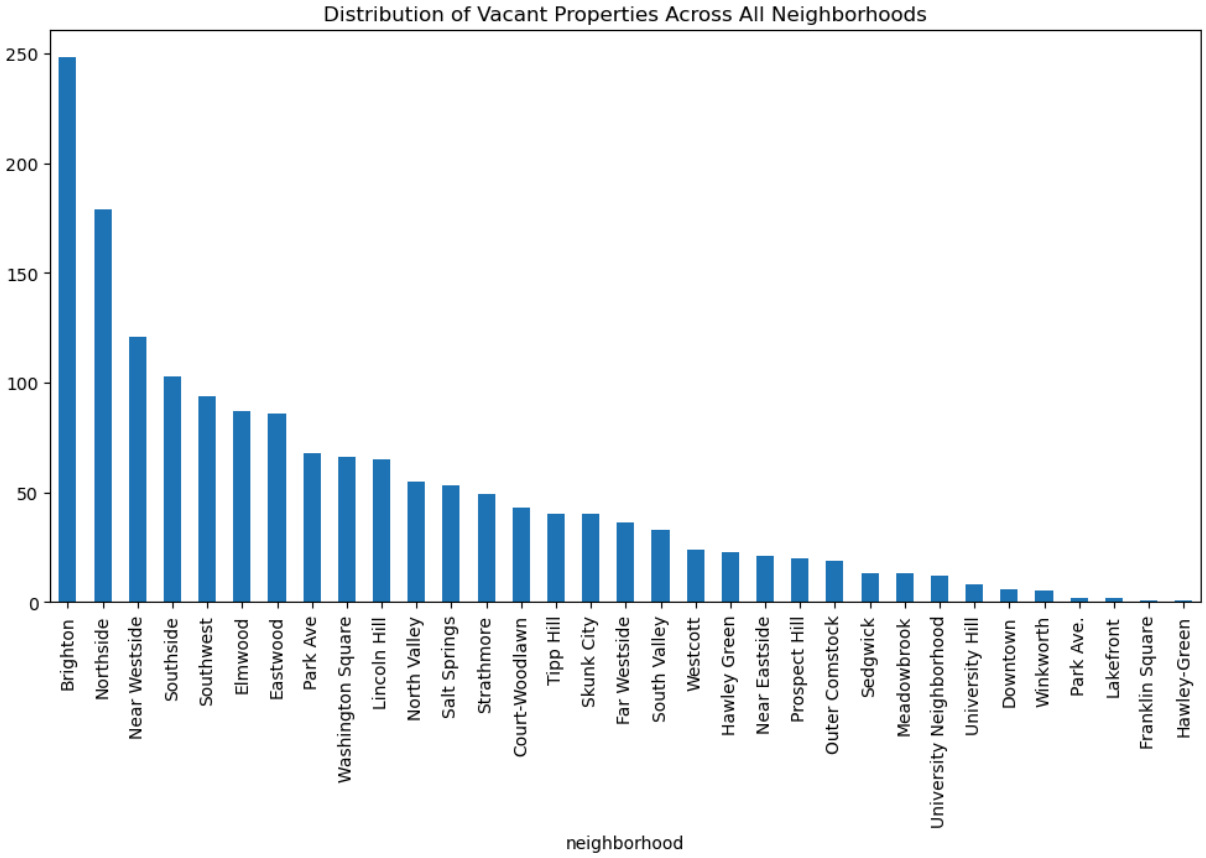


In [12]:
```
zip_counts.head(10).plot(kind="bar", figsize=(10,5))
plt.title("Top 10 ZIP Codes by Number of Vacant Properties")
plt.xlabel("ZIP Code")
plt.ylabel("Number of Vacant Properties")
plt.show()
```



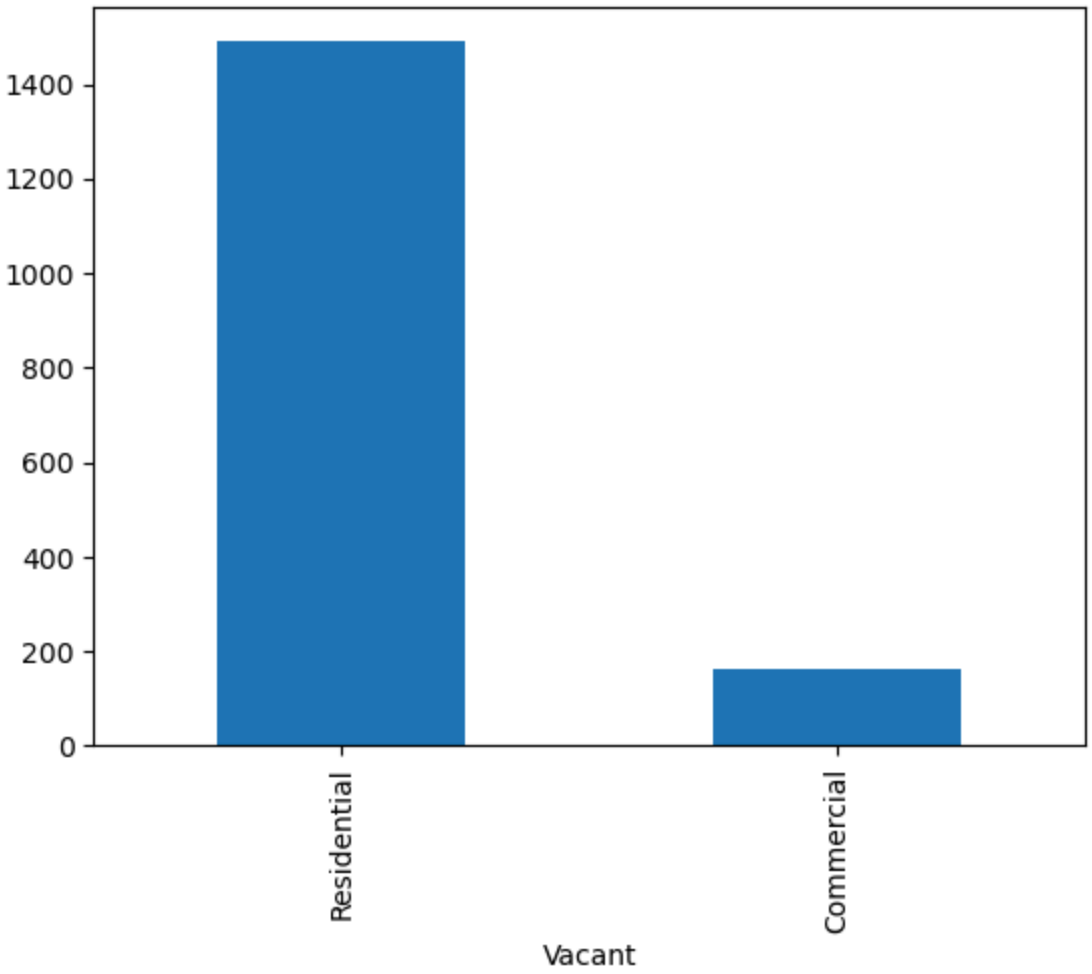
In [13]:
```
neighborhood_counts.plot(kind="bar", figsize=(12,6))
plt.title("Distribution of Vacant Properties Across All Neighborhoods")
```

```
plt.show()
```

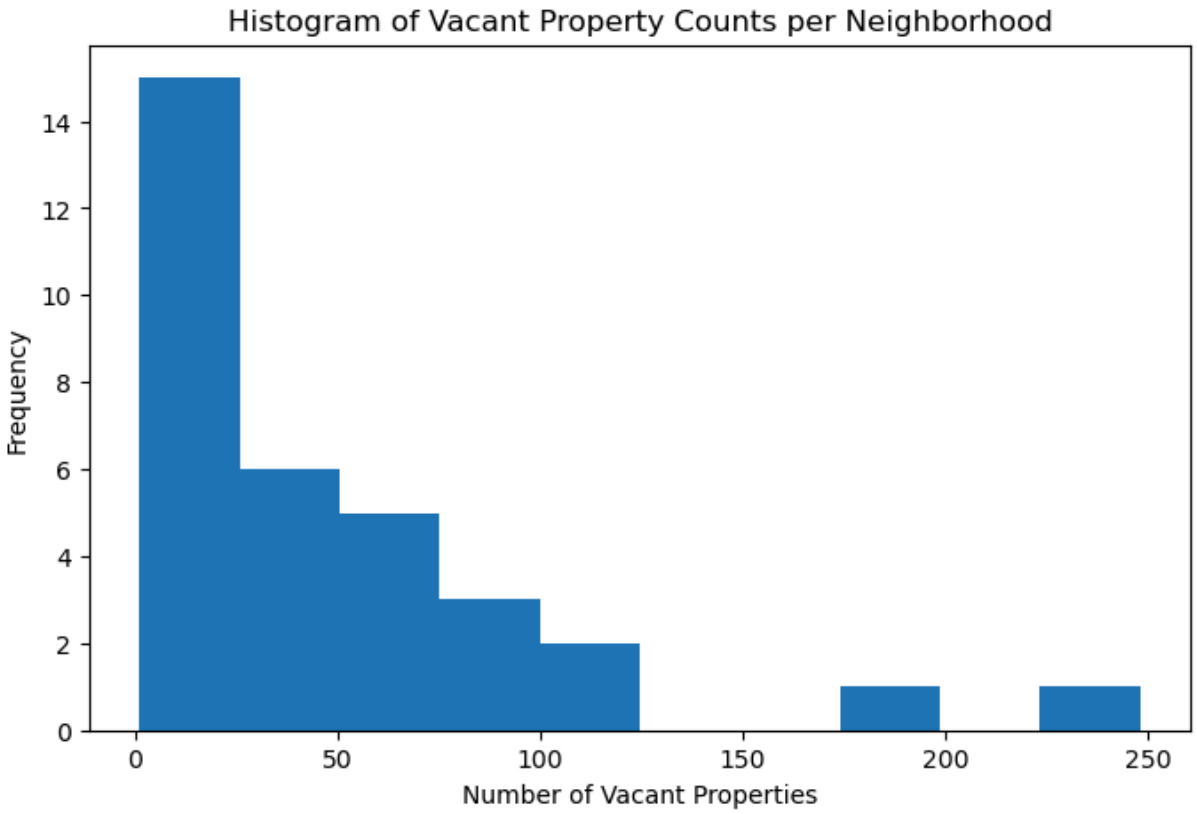


In [14]:
```
count_df["Vacant"].value_counts().plot(kind="bar")
```

Out[14]:
```
<Axes: xlabel='Vacant'>
```

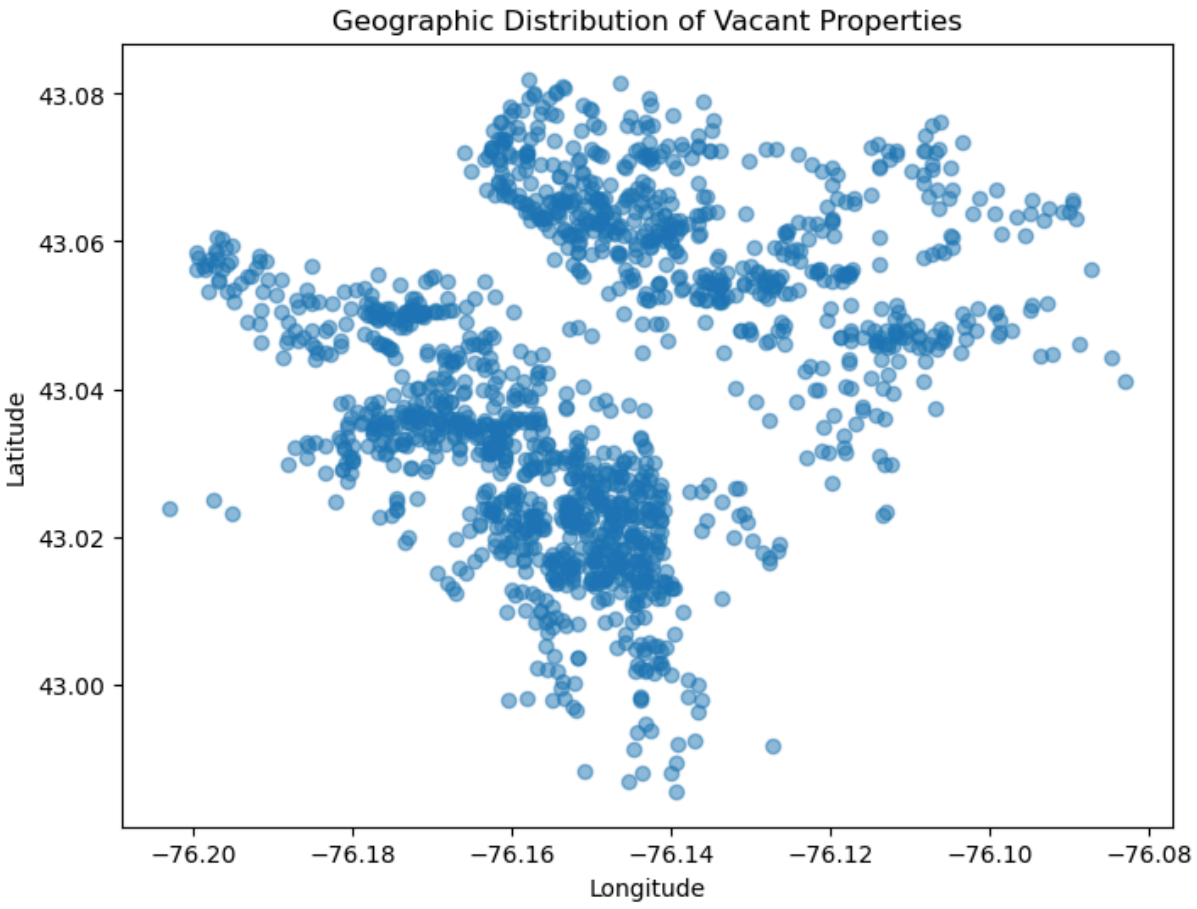


In [15]:
```
plt.figure(figsize=(8,5))
plt.hist(neighborhood_counts, bins=10)
plt.title("Histogram of Vacant Property Counts per Neighborhood")
plt.xlabel("Number of Vacant Properties")
plt.ylabel("Frequency")
plt.show()
```



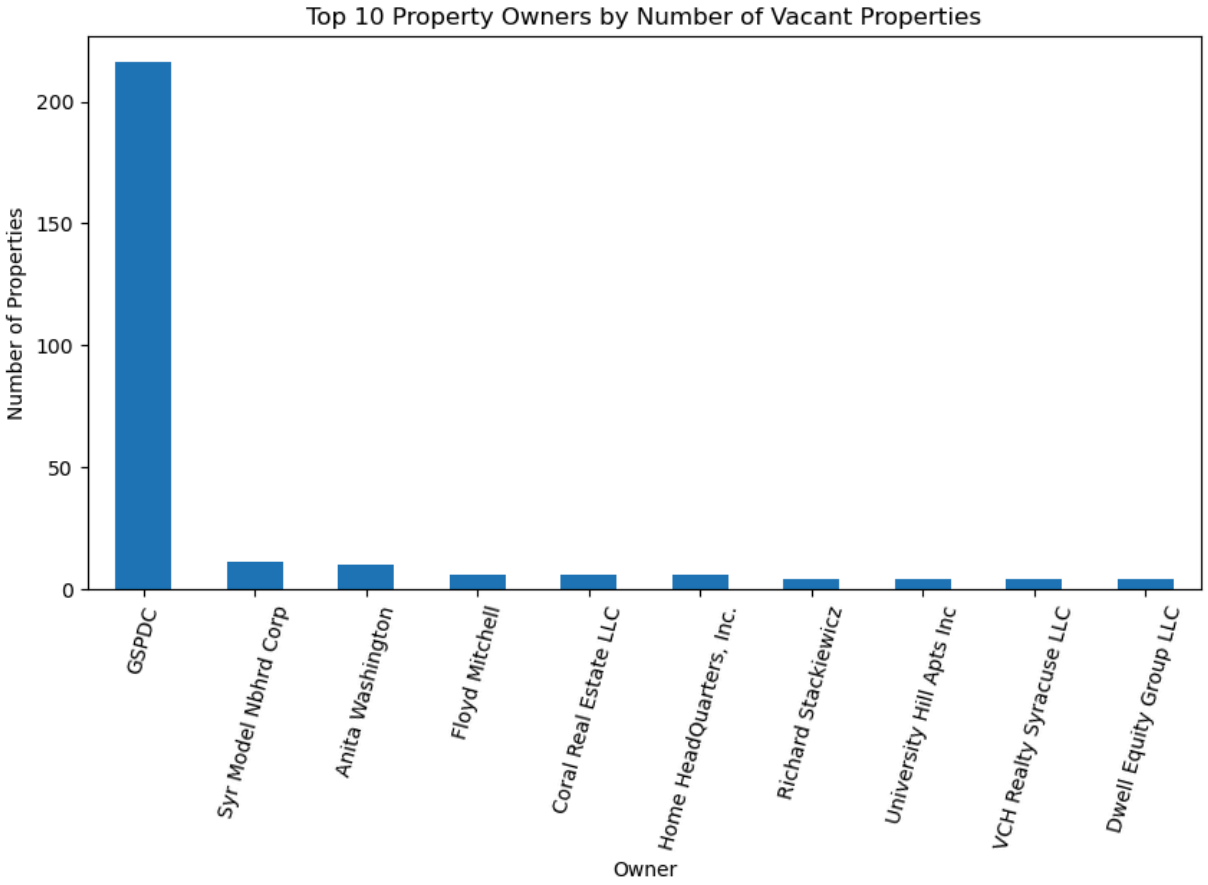
In [16]:
```
geo_df = analysis_df.dropna(subset=["Latitude", "Longitude"])

plt.figure(figsize=(8,6))
plt.scatter(geo_df["Longitude"], geo_df["Latitude"], alpha=0.5)
plt.title("Geographic Distribution of Vacant Properties")
plt.xlabel("Longitude")
plt.ylabel("Latitude")
plt.show()
```



In [18]:
```
top_owners = count_df["Owner"].value_counts().head(10)

plt.figure(figsize=(10,5))
top_owners.plot(kind="bar")
plt.title("Top 10 Property Owners by Number of Vacant Properties")
plt.xlabel("Owner")
plt.ylabel("Number of Properties")
plt.xticks(rotation=75)
plt.show()
```


Top 10 Property Owners by Number of Vacant Properties
Number of Properties
0
50
100
150
200
GSPDC
Syr Model Nbhrd Corp
Anita Washington
Floyd Mitchell
Coral Real Estate LLC
Home HeadQuarters, Inc.
Richard Stackiewicz
University Hill Apts Inc
VCH Realty Syracuse LLC
Dwell Equity Group LLC
Owner

In [ ]:

In [ ]: